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Transformative Business Studies - Technology Transfer in the Social Sciences

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1. Introduction

University technology transfer can be described as an organizational activity that transfers research results in the form of innovations or other knowledge that can be utilized for the benefit of society. Most commonly, it is accomplished through licensing intellectual property to commercial actors that are able to utilize the results in practical applications (1). This is a rather well-established process for transfer of concrete technologies, but research results that are more broad, such as those concerning organizational and social change, face challenges in undergoing the same process. Simply put, not all research results can be patented and easily replicated in different industrial applications. Nevertheless, the significance of the diffusion of academic thought grows as society addresses complex industrial and social challenges. This is the case for research results and solutions developed in academia that tackle a number of sustainability challenges, for example, within energy and logistics domains. The main question we ask in this paper is how business research should be organized and conducted so that it leads to meaningful change of behaviors. This is a crucial question since not any research will suffice. We thereby stipulate that the success of technology (and knowledge) transfer directly depends on the quality of research results produced and research process used.

Research in general is about producing new knowledge. New knowledge means knowledge that did not exist before. But for research results to lead to technology transfer the results also need to have a practical utility, they must in some way benefit society (or a member of society). If the result lacks practical meaning then it cannot be transferred, it has limited practical benefit. Thus, for a successful technology transfer there is a need to ensure that research results are relevant and valid and, at the same time, replicable in new contexts.

We argue that the research results that are 'transferrable' need to be produced within a pragmatic or 'actionable' research paradigm, which we refer as to 'transformative business studies'. In this paper, we define transformative business studies as research within business and social studies that is induced by real life challenges and leads to changes in business practices and larger social change besides producing general theoretical knowledge on the phenomena in question. We further stipulate that certain research modes are more conducive for performing such studies. In particular, *design-oriented research,* such as design science research in organizational studies (2–5), intentionally produces results that can solve real-life problems rather than generate only explanatory or descriptive knowledge. *Participative research* methods (6–11), on the other hand, facilitate ensuring the relevance and validity of produced knowledge. Finally, stabilizing the actors supporting or opposing a solution, inspired by *Actor-network theory* (12–16), is a crucial part of successful technology transfer. The combination of these research methodologies and paradigms,

we believe, forms a strong methodological basis for producing 'transferrable' research results that concern larger organizational and social change, and thereby are particularly suitable for performing transformative business studies.

To illustrate the research process that we propose for successful transformative business studies, we utilize an example of a method for designing business ecosystems that was developed in a series of projects. Biogas-for-traffic business was the context for these projects, and their main practical aim was to design a business ecosystem that would enable utilization of biogas as fuel in public and commercial transportation in a focal country. The theoretical aim of this study was, in turn, to develop a method for designing such enabling business ecosystems that could be applied in other similar problematic contexts, such as introducing cleaner technologies in the somewhat rigid and institutionalized energy and logistics industries.

In next sections, we describe the methodology for performing transformative business studies that is based on design science research, participative research and actor-network theory, which is followed by the example of how it was applied in the biogas-for-traffic case and which concrete research tools were used. We further discuss the transferability of research results produced in the exemplary research case and conclude by outlining implications for researchers and practitioners.

2. Methodology for transformative business studies

2.1. Design science research

Producing transferrable research results requires a research process with transferability as its objective. Research in general can be classified in two different modes, reductionist analysis and purely descriptive. The former is found primarily in the natural sciences taking "nature apart to discern and understand its fundamental units" (25), and the latter is common for sciences describing social and cultural phenomena, such as humanities. There is, however, the third mode of research, which is not always perceived as 'pure science' due to its applied nature – design science research (2,25,26,20). Design science research (DSR) is the underlying paradigm in such disciplines as engineering, medicine, and computer science, because its aim is to create knowledge that is prescriptive and that leads to action with improved results. While DSR is a natural research mode for the named sciences, the research results that can foster greater sustainability improvements are prone to include more than only engineering; a certain degree of process, organizational and social engineering (influencing attitudes) is necessary. Organizational and social sciences, however, are far less familiar with design science, research though attempts to adopt this mode of research have been made (2,26,20).

DSR, as opposed to explanatory and descriptive modes of research, uses target systems and (design propositions) when defining the initial situation and urges people towards thinking how the system could be made to work (3). The main outcomes of such research include the design proposition, i.e. the desired outcome, and the knowledge on how to achieve it (3) that can be transferrable to other similar contexts. The truth of such knowledge can be assessed in pragmatic sense, that is, as James puts it "its verity *is* in fact an event, a process: the process namely of its verifying itself, its veri-*fication*." (32). Knowledge of this sort is able to support and motivate practitioners for actions in order to achieve the desired goal.

This is in contrast to purely explanatory knowledge, which is able to answer the question 'how to' only by establishing general cause and effect rules or laws, and extrapolating into the future. The

difference between the three main modes of research and the knowledge produced within them is demonstrated in Table 1.

Table 1 Comparison of the three modes of research (Adapted from (20)*; based on* (3)*)*

	Science	**Humanities**	**Design**
Purpose	Understand phenomena by uncovering causalities, patterns and forces that underlie these phenomena	Describe, understand and critically reflect on the human experience of phenomena	Produce the systems that do not exist yet, i.e. change the existing situations into desired ones
View of knowledge	Representational: - our knowledge represents the world as it is; - nature of thinking is descriptive and analytic	Constructivist and narrative: - all the knowledge arises from what actors think and say about the world; - nature of thinking is critical and reflexive	Pragmatic: - knowledge is in service of action; - nature of thinking is normative and synthetic
Character of knowledge	Reductive analysis	Descriptive	Prescriptive (actionable)
Examples of scientific domains	Natural science (physics, mathematics, biology)	Humanities (history, hermeneutics, literature)	Design and engineering based upon diagnosis (engineering, medicine, architecture, computer science)
Focus of theory development	Discovery of general causal relationships among variables (expressed in hypothetical statements): Is the hypothesis valid? Conclusions stay within the boundaries of the analysis	Key question is whether a certain (category of) human experience(s) in an organizational setting is “good”, “fair”, etc.	Does an integrated set of design propositions work in a certain ill-defined situation? The design and development of new artefact tends to move outside boundaries of initial definition of the situation
Metaphor	Life understood retrospectively		Life as it can be

DSR is relevant for addressing research problems, where the underlying laws are not discoverable, too complex to be reduced into a limited number of variables, or are even irrelevant (e.g. due to their high degree of conditionality). Sustainability challenges, being an example of social challenges that require transformative business studies, fall under this definition due to their complexity. Academia is responsible for producing the research results that are not only of descriptive and explanatory character, but also, and importantly, applicable, ‘actionable’ knowledge that can put humanity on a path to sustainable development.

2.2. Participative research

The second corner stone of the methodology for transformative business studies is the participative action research (PAR) mode, which can ensure the relevance and applicability of research results, and thereby their ability to transform the society. There is a number of methodologies associated with PAR, such as action research or clinical research. In this paper, we specifically focus on clinical research (8,38,40–45) as the methodology for performing PAR due to its focus on extant involvement of organizations as active research participants and refer to it as PAR or clinical research or inquiry interchangeably. It builds on the idea of engaging in research activities that are based on the needs of organizations and the co-creation of solutions in collaboration with said organizations. It allows not only the accessing of sensitive contexts, but also ensures commitment of the research participants to co-create valid knowledge (9,39). The validity and, what is equally important, the relevance and actionability of the produced knowledge are ensured by studying social systems 'as they react to experimental manipulation' (46).

The sufficient involvement of relevant practitioners, we believe, is necessary due to the following reasons:

- *Complex and ill-defined research problems*: when the target of the research is not clear, there is not only the question of 'how to solve the problem?' but also that of 'what is the problem?', and the dialogue with practitioners helps obtain a multitude of angles on the problematic context.
- *Need for change*: when the purpose of the study is to trigger change in an industrial organization, the involvement of the actors who are to carry out or will be affected by the change is necessary in order to ensure its success.
- *Institutionalization*: problematic contexts which require transformative business studies are prone to institutional rigidity, uncertainty, organizational inertia and lack of cooperation. Thus, the change of attitudes and ways of working can be better achieved when the research is done in cooperation with the key actors, rather than through 'handing over' the research results to the practitioners for implementation after the study is over.

The involvement of practitioners can be achieved through employing both consultants and researchers in the study as demonstrated in Figure 1. The role of the researcher in this process is to understand each of the problems inside the context (48) and together with the consultant generate actionable knowledge. The role of the consultant is to ensure the benefit of the client and those stakeholders which are deemed relevant for achieving a sustainable outcome. The researchers and the consultants set the stage for problem solving (11) as the context evolves, and new problems emerge. This enables the practitioners to explore and act upon the emerging knowledge thus generating actionable data (38).

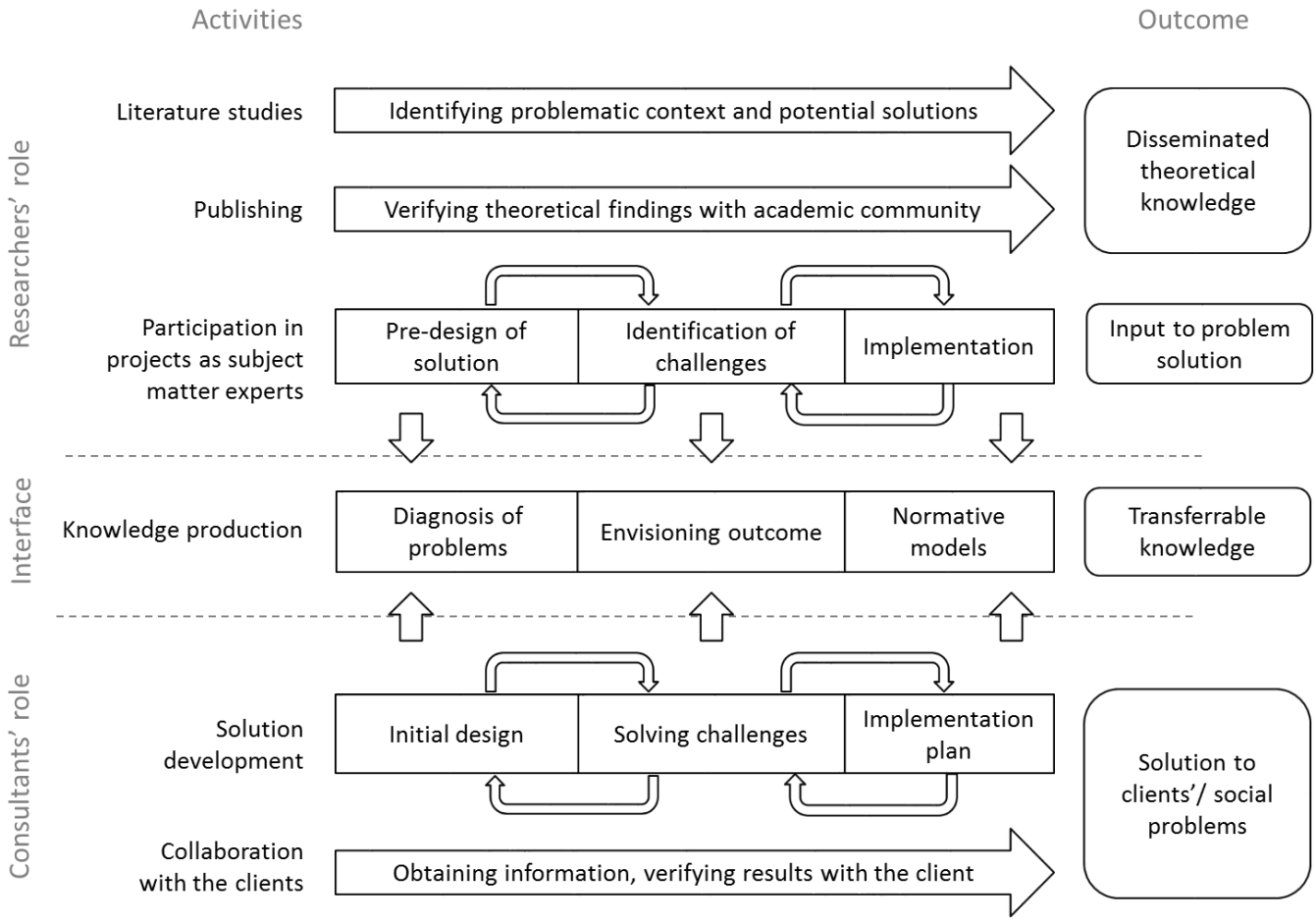


Figure 1 Organization of a transformative business study and the role of researchers and consultants

The collaboration between researchers and consultants gives researchers access to rich data while giving clients access to new knowledge that is developed and applied with the aim of improving the client's situation. The iterative process of research and application at the same time allows for abstraction of research results so that the knowledge can be applied in other contexts so that the benefit to society and the relevant stakeholders is realized. Though practitioners have knowledge of the incumbent context, the role of the consultants and the researchers is central as new knowledge emerges and the project moves, according to plan, into unknown territory.

The researchers here draw upon more general, fundamental sources to bring insight into the emerging knowledge landscape. This requires self-awareness and reflection on the part of the researcher (7). The process is driven by a need for relevance in terms of actionability which makes the knowledge valid in a pragmatic sense (32). Through participatory research methodology, i.e. clinical inquiry, the researchers, consultants and practitioners create knowledge that at the same time is actionable and can be generalized. The validity and relevance of this knowledge is measured in its actionability. The fact that the solution works is proof in itself and the more it works the truer it is.

2.3. Actor-network theory and controversy mapping

The development of the design proposition for a complex research problem can be equally dependent on economic, social, technical, physical, and political features, and other factors. To

design a solution, none of these critical issues can be left out, because it might undermine the resilience of the solution as such. The design proposition is chosen based on the type of theory which is best applicable to the model. In case no applicable theory is available the problem is approached inductively focusing on why the theories do not solve the problem. Moreover, it is difficult to prescribe the success of developing a solution based upon one set of factors. At the point of transfer, the knowledge is part of a complex socio-technical actor-network which together forms the solution (12,35–37).

To fully understand the (social) requirements of the solution, that is, one needs to understand the complex structure of elements that influence and form the problem at hand. One tool for understanding the actor-network in any particular context is to identify and map the controversies and the combination of programs and anti-programs forming the controversies (14). The purpose of controversy mapping is to identify the arguments supporting and opposing a system or a solution. This helps understanding how the opposing arguments (anti-program) and who holds them can be co-opted, countered or eliminated and how the supporting arguments (program) and who holds them can be strengthened. The front line of controversy demonstrates whether the program or anti-program is prevailing and how the alignment of forces changes over time. As the issues are tackled iteratively the controversy front line between the program (p) and anti-program (ap) will shift (see Figure 2). In order to achieve a functioning solution, the researchers and consultants need to be able to enroll enough arguments (p) on their side, strengthen these arguments, and weaken and co-opt the opposing arguments (ap).

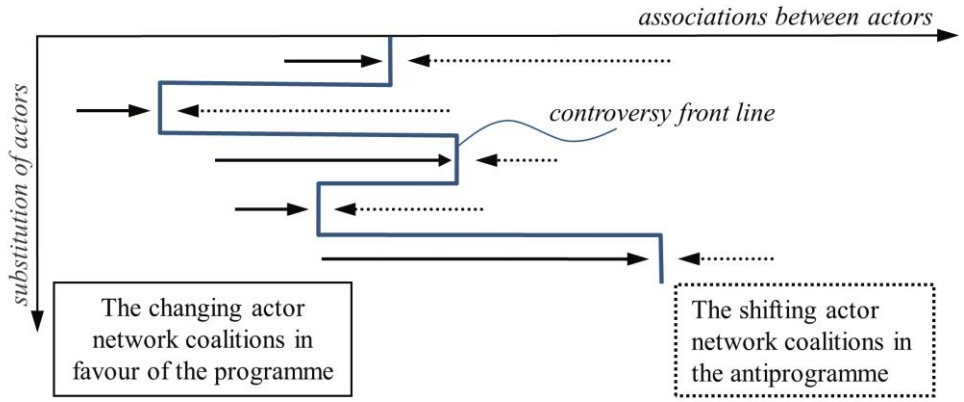


Figure 2 Exemplary controversy mapping diagram

Controversy mapping is one tool that, as we propose, can be used for addressing the actor-network related to a solution developed within transformative business studies. It is deeply intertwined with the other two cornerstones of the methodology we propose in this paper, because it aims at helping reach a good and 'implementable' design proposition while rooting it with the practitioners who are to implement, affect or are affected by the solution.

3. Exemplary Ecosystem Design Research Project

The method for designing business ecosystems, used as an example of transferrable knowledge in this paper, was developed within a series of projects that, we argue, were carried out as transformative business studies. The projects were organized as collaboration between academia (Åbo Akademi University) (17) and a research-based consultancy (PBI Research Institute) (18).

PBI Research Institute (PBI) is a research-based management consulting company that collaborates with Åbo Akademi University (ÅAU) to develop new knowledge which is then applied in practice. PBI focuses on industrial investments and projects especially in the transportation and energy sectors. PBI strives to adopt, transform, create and implement new knowledge on how management practices may be improved in order to enable more sustainable investments and projects. Sustainability can be broadly understood as meeting the needs of our generation without compromising the ability of future generations to meet their needs (19).

In this format, research and consulting is done hand-in-hand. Whereas the consultants can be considered results-oriented general practitioners, researchers participated in consulting projects as subject matter experts. As consulting projects are led by consultants, it is the project manager who choses which researcher to engage in the project. Researchers and consultants also engage in a continuous dialogue with researchers querying the consultants as to how a certain phenomenon looks in practice and consultants, for example, asking researchers to elaborate on a persistent phenomenon they have observed. Consultants also apply the concepts previously developed with the outcome fed back to research. Thus, the collaboration between researchers and consultants is continuous and iterative in the PBI- ÅAU arrangement.

The common feature of the projects that we use as an example in this paper was the interest in business ecosystems in which the respective actors operate or could operate. Besides solving actor-specific problems and producing respective research results, the main scientific outcome of the projects was the method for designing business ecosystems. We argue that the research mode and methodologies utilized during the research projects allowed us to develop and transfer such research results that were for practical use despite the complexity and uncertainty associated with the design of business ecosystems. While part of the research results were case and context-specific, the method and mindset for developing business ecosystems were formulated on a more generic level so that it would be applied in multiple contexts. As will be demonstrated, the method was later applied in other industrial contexts. It is important to note that the application of the research results in practice in other cases, i.e. the technology transfer, did not stop the research process; the method for designing business ecosystems was further developed and new aspects of the problem, as well as solution, were discovered.

Specifically, the series of projects related to the usage of biogas in transport (20–24). Biogas is methane that arises from anaerobic digestion of biomasses such as sewage sludge, manure, household waste or crops. Biogas can be produced as a by-product of the sewage treatment process. Biogas is a locally produced renewable fuel that reduces local imports of fuel and refines local waste into transport fuel. The technologies for using biogas in transportation are commercially available. Biogas is used to a large extent in Sweden, especially in public transport, waste handling and in passenger cars. However, biogas was not used in transportation in Finland which is Sweden's neighbor. The goal of the project was thereby to establish biogas for transportation in a municipality in Finland and then expand it to other parts of the country. The research studied the challenges of 'green' technologies with local economic benefits and their adoption in Finland. In doing so the project team worked together with multiple actors in the ecosystem (see Table 2).

Table 2 Ecosystem actors involved in the exemplary ecosystem design project

Role in the ecosystem	**Ecosystem actors**

Biogas production and distribution	• Biogas producers • Biogas and natural gas distributors • Other fuel distributing companies
Biomass supply	• Farmers and their representatives • Waste management companies (including wastewater treatment)
Biogas consumption	• Fuel users (transportation, delivery, waste management companies)
Equipment supply and maintenance	• Suppliers of gas-driven vehicles • Equipment and facility suppliers (biogas treatment, distribution) • Engineering companies
Policy-making	• Various departments of focal municipality authorities (environmental, transport, waste management), which represented the citizens • Relevant state-level governmental and non-governmental authorities
Financing	• Funding and investment organizations
Research and benchmarking	• Research organizations (research in farming, transportation, etc.) • Companies and municipalities that consume biogas in Sweden (for benchmarking and experience sharing purposes)

Throughout the process, as the team from a consulting point of view tried to solve the problems with adopting biogas as traffic fuel, the researchers in the team were at the same time making observations about what kind of challenges the combined project team (researchers and consultants) was facing.

4. The application of methodology for transformative business studies in the exemplary project

While no particular research question was initially identified, the biogas-for-traffic project started with the general question why 'green', potentially sustainable technologies such as biogas production from waste and by-products of different industries do not get traction in the market, and, particularly, in the Finnish transportation sector. The specific interest was in understanding how an enabling business ecosystem could be formed. The choice was between studying existing ecosystems retrospectively and inducing how they were created or participating in actually creating them. The main reason for choosing DSR over historic, explanatory or descriptive research was the need for producing prescriptive, 'actionable' knowledge, i.e. knowledge on how to tackle problems generated from the process of tackling the problems and recording both successes and failures. Moreover, existing retrospective studies on successful development of industrial eco-parks shed light on the success stories, but were deeply rooted in local contexts (27–30). In practice, most of such eco-parks appeared spontaneously (27,31), and thus the motivations and fine details of their development could be missed or simply forgotten when the descriptive research about their organizational evolution was undertaken.

In the focal case, initially no concrete problem, such as e.g. lack of political will or lack of technological innovation, could be pinpointed as a reason for struggles with developing a biogas-for-traffic solution in the country. The answer for the question could not be found from prior studies, because the question was yet unknown. The problem needed to be found 'on-site', which justified the need for the researchers to be part of the team of consultants developing a new ecosystem in order to identify pitfalls and hindrances as they occur real time.

Another reason for choosing the DSR approach was its emphasis on a creative leap between knowledge that is generated in exploratory and explanatory science and a design proposition (2). The development of a design proposition is not based on the causality discovered in 'pure' science (A causes B, therefore do A if you want to reach B), but is rather based on pragmatic 'reasonableness' of a solution (C is the problematic context, D is a design proposal based on the vision of an ideal target outcome O; if D overcomes problems C and leads to O, then it is a good solution). A proper solution to a complex and ill-defined problem cannot be induced from a problem or deduced from theory, but can only be *informed* by problem specification and theory. Thus, theories in this case serve as instruments to explore the problematic context and devise means to change it (32).

The incorporation of DSR into the research methodology for transformative business studies was the first step for ensuring that the knowledge produced at ÅAU is actionable at PBI and for its customers.

To achieve the relevance of the study, the biogas-for-traffic projects relied on a clinical inquiry. For this, both researchers from ÅAU and consultants from PBI were involved the project. In certain cases, the same people acted both as researchers and consultants, but the roles and responsibilities differed accordingly. Importantly, the consultants' role was to keep an ongoing dialogue with the practitioners in order to diagnose the research problem and ensure the practical relevance of the produced solutions to this problem.

Following the process presented earlier in Figure 1, the responsibility of researchers was in providing theoretical insights while diagnosing the problem and developing solutions together with the consultants, thereby acting as subject matter experts. In the focal case, their role was to bring the existing body of knowledge about business ecosystems and biogas-for-traffic business into the research process. The consultants, on the other hand, were responsible for developing the working solution that would solve the clients' problems related to using biogas as traffic fuel. For that, consultants built a direct interface with the clients through actively involving them in the project meetings, interviews and workshops. This was necessary for both obtaining the first-hand information about the challenges and possibilities in biogas-for-traffic business and ensuring that the produced knowledge is useful for solving the clients' problems.

The interface between the research and consultancy was the actual knowledge production achieved through joint project participation. While the consultants verified the actionability and usefulness of the produced knowledge with the practitioners, the researchers' responsibility was to verify the academic merits of this knowledge, i.e. its originality and ability to be applied in other contexts. This allowed to achieve both actionability and transferability of the produced knowledge.

To provide an example of how the knowledge produced had both practical and theoretical merits, during the research within the biogas-for-traffic case it appeared that the scope of local delivery trucks needed to be redefined. Local delivery trucks are usually bought primarily as a unit, not a lifecycle solution since diesel is a homogenous abundantly available product, and there are many competing service centers. A biogas-driven truck would be dependent on a single supplier with uncertain service support. To address this issue, it was decided that biogas delivery trucks would be offered including lifecycle service support and fuel. Customers would therefore pay a certain price per kilometer and would not be exposed to fluctuations in the fuel price. This was possible because biogas production and price was not strongly influenced by fluctuations in the oil price. The offer

was limited to customers who signed up for leasing large enough fleets of biogas-driven trucks for long enough time, but would allow pooling among customers to aggregate demand. Ensuring a large-enough fleet in turn was a prerequisite for the truck provider to invest in the local service center. In this example, the generated context-specific knowledge included the understanding that in Finland, a total package including the vehicles, their servicing and fuel would be most attractive for involving various truck operators in the biogas-for-traffic ecosystem. More generalized knowledge, which could be transferred to new contexts, included the solution mindset, which aims at tackling potential customers' uncertainty related to the introduction of new technology to the market.

Finally, controversy mapping was used as a tool for stabilizing the actor-network related to the business ecosystem in focus. Since the problem of enabling biogas-for-traffic business was complex, ill-defined and multi-variate, one strength (p) could easily be countered by a weakness (ap) in another dimension. For the solution to function, it had to be viable, it had to work economically, environmentally and socially; otherwise it would face opposition in whatever dimension it falls short. This meant the boundaries of the solution were not static. As new knowledge and problems emerged, new arguments were needed in order to stabilize the context and control the anti-programs that may hinder the solution. Stabilizing here means making sure that there are no uncontrolled issues that may introduce uncertainties that could make the entire solution collapse.

In the biogas-for-traffic case, it meant, for example, engaging farmers in order to be able to ensure reliable raw material supply for biogas production (p). Although initially waste management companies were considered as the main biomass suppliers (p), analysis showed this raw material flow would not be sufficient to scale up (ap). This was a problem. For the solution to be economically sustainable, it had to be scalable (ap). Therefore, an additional source of raw material had to be found that could meet anticipated demand. Such raw materials can be grown (p) but doing so would displace cash crops driving up costs to unsustainable levels (ap). The solution to this was found by growing biogas-appropriate crops simultaneously (p) in addition to the traditional cash crops (p). These crops were traditionally not considered valuable but would bring revenues in this case (p) and could be grown without reducing cash crop acreage (p), thereby reducing the cost pressure on farmers. This action would also improve the soil composition and thus support the main cash crop (p).

As parts of the solution become actionable, these were 'blackboxed' as plausible actors. Figure 3 shows the blackboxed sub-systems, or modules, of the biogas-for-traffic case. For example, module 3 'Biogas for truck operators' can be separated from module 4 'Biogas for car users' because the two modules are functionally different. Trucks are high capex, consume large amounts of fuel and are purchased based on a business-investment calculation. Module 2 differs from trucks in the sense that the municipality has a very strong say since the investment is based on municipal tendering of bus services. Plausibility does not mean this was the best or ultimate solution, only that it was good enough under the circumstances in this context. At the same time, those actors that are unresolved were taken apart in order to address the parts of the problem and thereby strengthen the program and enroll new actors to counter the anti-program.

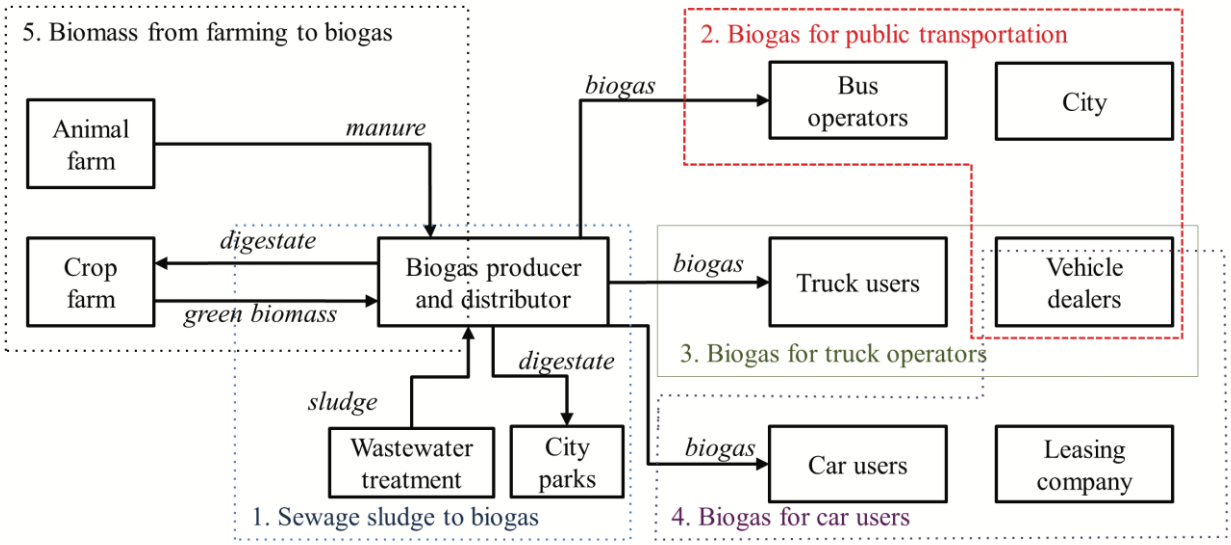


Figure 3 Blackboxed subsystems (modules) in the target ecosystem

One example of taking a problem apart and blackboxing its elements was the question of what kind of pricing mechanism would be suitable for biogas sales. To facilitate the investment in biogas production and distribution infrastructure, the pricing logic was of crucial importance. Low prices would be good to lure customers (p) but would make scaling up difficult (ap) and would make it more difficult to get financing for the investment (ap). The solution was in dividing customers into different categories. Big customers that commit early to the biogas-for-traffic ecosystem would get long-term contracts with fixed (inflation indexed) prices. This was motivated since the customers were taking a CAPEX risk (ap) committing to technology that was more expensive (ap) and with unclear second-hand value (ap). It was possible since biogas production was not exposed to variations in the oil price (p). For small customers, the price would be allowed to vary more in line with oil prices (petrol and diesel) which would provide a healthy margin (p). Table 3 shows the characteristics of subsystem 3: the current business logic of each actor, the risks and uncertainties related to joining the biogas-for-traffic ecosystem and the change in business model needed for each specific actor. The integrator, in this case the biogas producer and distributor, is a special actor that integrates all the other actors into the ecosystem.

Table 3 Characteristics of subsystem 3 Biogas for public transportation

Actor	Current business logic	Risks and uncertainties related to joining ecosystem	Change in the business model
Truck operators	Fuel is a significant operational cost Green image is important Drive locally but can transfer vehicles to other cities where they operate	Investment into gas-driven trucks is larger than into diesel trucks Uncertainty about biogas technology: maintenance costs and reliability Uncertainty about fuel: price and availability of biogas Second-hand risk of owning gas trucks Limitations of operating buses to the locations where gas is available	Switch of fleet to gas-driven trucks completely of partly Possible: charging more for the 'green service'
Truck dealers	Gas-driven trucks are not sold in the focal municipality due to the absence of fuel	Investment into gas truck maintenance infrastructure is feasible only if large number of trucks is serviced, i.e. is sold in the first place	Sales of trucks together with maintenance agreement, leasing option and long-term fuel guarantee from the biogas distributor
Integrator		Investment into distribution infrastructure will not be recouped if high enough consumption volumes are not ensured	Long-term contracts for biogas supply at fixed price Low-emission (CO2 and NOx), locally produced fuel compared to diesel

Thus, actionable knowledge is not generated only as the ultimate outcome. It is constantly produced (and sometimes overturned) throughout the research process. The actual end-point of the research is reached when all the key parts of the system have been stabilized to the point that the practitioners are able to proceed by taking action according to the knowledge generated (11,38,39). Therefore the outcome of transformative business studies is both knowledge and actual change that can be generalized from the specific case (7,38).

5. Transferability of research results

The biogas-for-traffic projects resulted in specific recommendations as to how the municipality, in collaboration with the key stakeholders, should proceed. This result is context-specific and thus cannot be easily replicated. The projects also resulted in a generic academic model for how complex business ecosystems could be mapped and redesigned alongside the practical, context-specific knowledge. It consists of the following sequence of specific steps:

1. Structure the business ecosystem according to functional modules (figure 3)
2. Establish collaboration between the modules with incentivizing mechanisms
3. Enable scaling of the ecosystem through replication based on modularity
4. Integrate the functional modules with the help of boundary spanning business models which build on the incentives in the adjacent modules

Considering a business ecosystem to be a system in general, it can be seen as a set of interconnected functional modules that together produce a system output and certain value to the end-user or society. Depending on how well they are connected and fit through interfaces, and what kind of elements fulfill the functions, the output is produced in a more or less efficient manner. Companies, and or market actors, by employing their business models and technologies, fulfill the functions. The structure of current business ecosystems can be analyzed, and a new, improved structure can be proposed based on the analysis of inefficiencies and missing interfaces between functional

modules. Most often, the boundaries of the modules change, i.e. new actors are introduced, others are removed, and new interfaces between the companies are needed.

In the biogas-for-traffic case, such modules included, for example, a number of biogas consumption modules (step 1): 'biogas for public transportation', 'biogas for delivery business', 'biogas for individual users'; a number of biomass supply modules were also identified (step 1): 'biomass from farming' and 'biomass from waste management'. Each module had a specific business logic, including value creation logic and benefit sharing between participants (step 2), which made it distinct from another. For instance, while biomass obtained from farming would be of good quality but would cost (step 2), the biomass from waste management would have a negative price (step 2). Thus, the approach towards incentivizing respective actors to become biomass suppliers in the biogas-for-traffic ecosystem would differ.

The development of incentivizing mechanisms is the next step which aims at attracting the critical actors in the business ecosystem. Based on the logic of each functional module (step 1), these mechanisms are designed (step 2). The research done in the biogas-for-traffic case allowed not only to develop context-specific mechanisms, but also formulate more generalized and thereby transferrable principles of enrolling actors in business ecosystems. The example of providing a lifecycle solution instead of just a technological product to customers in the face of uncertainty (discussed in section **Error! Reference source not found.**) is one of such mechanisms. Development of an alliance of critical actors that can fairly share the benefits of the total ecosystem working more efficiently is another example of an incentivizing mechanism (step 2), which was later applied in a short sea shipping ecosystem development project (see case 3 below).

Due to modular design (step 1), it is possible to replicate the business ecosystems. In the biogas-for-traffic case, for example, it proved to be necessary to extend the biogas-for-traffic business to other cities in Finland so that a reliable and convenient fueling network could be formed. For this, the cooperation models between the biogas company, biomass suppliers and biogas consumers (step 2) could be conceptually copied (step 3) in those locations with the necessary adjustments (step 1), i.e. replicated. This is possible due to the form in which actionable knowledge produced within DSR is recorded: the solution is described together with the problems which it solves and mechanisms that make this solution work for these specific problems. Then, in new contexts, it is possible to identify which problems actually need to be solved and thus which incentivizing mechanisms need to be applied in order to align the incentives of the stakeholders.

The final step is the development of a boundary-spanning business model that is based on the results of the previous steps. This is especially relevant for companies that attempt to become 'ecosystem integrators', i.e. the ones that integrate the whole business ecosystem and through various incentivizing mechanisms change the business models of other actors (step 4). It is not always clear who would become such an integrator, and in subsequent projects it was often the case that a consortium of interested actors would form such a strong player who would be able to alter the ways of working and value creation structure of a whole business ecosystem.

This method for designing business ecosystems has been an is being used successfully in a number of consulting projects. It has proven especially suitable in two types of situations, enabling a new technology to enter the market and changing the earning logic of an industry so that it becomes more productive (partly by taking into use new technologies).

Case 1 Unmanned ships: Unmanned ships that could significantly lower freight costs and emissions as well as reduce accidents since sea freight is an accident-prone industry. However, the incumbent business models in sea freight do not enable a profitable investment in unmanned ships. To achieve that the business ecosystem was redesigned, redefining the business models for key stakeholders and how by coordinating the strategies the new earning logic of the ecosystem could be implemented.

Case 2 Flexible power generation: The increase in wind and solar power has changed the earning logic of the energy systems putting a premium on cost-efficient flexibility instead of high-efficiency continuous load (40). However, the incumbent business models and especially investment models in power generation made an investment into flexible technology unprofitable. The business ecosystem and specifically the investment model was therefore redesigned. Specifically, a new division of roles in the investment structure was established which enabled a better allocation of risks and rewards and a more efficient employment of capital. The solution, enables the introduction of technology that reduces system emissions. The need for state subsidies is also reduced since such a division of value creation and capturing is established that all the key actors required for a functioning solution get a share of the improved performance on system level.

Case 3 Short sea shipping: Short sea shipping is today highly inefficient in terms of current market condition capacity utilization. The utilization rate of vessels is 35-40% due to long stays in port (40% of the time) and sailing empty (20-25% of the time). This results in low profitability and high freight costs. By redefining the division of roles and responsibilities in operative logistics planning and in the planning of infrastructure investments a more productive ecosystem with a higher, more effective utilization of the capital employed can be attained (the actual launch is taking place as of writing). Emissions also sank partly because of the higher utilization rate but also because the new division of roles and responsibilities enabled business models based on more efficient technologies and operations (49,50).

The continued application of the research methodology for transformative business studies described in this paper, which is based on DSR, participative research and ANT, benefits the companies to which it is applied. It also contributes to research by facilitating continued research on the subject and further refinement of the model. The collaboration between researchers and consultants is central here. The more the consultants apply the method, the more they will be able to point to weaknesses and possible improvements.

The key element is the focus on and creation of actionable knowledge. Redesigning an ecosystem does not mean publishing a report that only outlines how to carry out the change. The method includes the actual change by producing context-specific actionable knowledge since it is only when actually applying the recommendations, and the variation to the recommendation and so on, that you can say what knowledge it was that triggered action. That means knowledge that compels the key stakeholders in a business ecosystem to act in a way that brings about the new way of working, essentially changing the behaviors in the marketplace. The knowledge compels to act in a certain way much like the axioms laid out by Suh et al. (51) compel designers and managers to design products in a certain way because not doing so would be irrational.

The fact that the case described is closely connected with innovation is a coincidence. PBI in collaboration with ÅAU has previously created methods for managing trust in B2B relations and methods for pricing services based on their value added, to name a few (44,45). In both cases, the

research process developed according to the above steps so that the symptoms where studied, problem identified and subsequently a solution that was turned into a service (in these two cases a web-based service with analysis support).

6. Conclusion

We developed a method for designing business ecosystems, which addresses complex and uncertain phenomena. The challenge was to create and formulate the knowledge in such a manner that it is possible to transfer it to business application, i.e. make it a tool for solving similar challenges in the future. Given the complexity, uncertainty and dependence on the context when designing or transforming any business ecosystem, it was difficult to formulate clear and universally applicable rules. Instead, the knowledge was documented as 'actionable', i.e. not only the solution, but the identified problems that led to this solution were documented. The transformative business study undertaken by the researchers needed to be based on DSR, participatory research and actor-network-theory. Since the solution did not exist beforehand, we relied on DSR to outline the goal to be achieved. Participatory research allowed better access to most sensitive challenges that would otherwise be undiscoverable. This is because organizations are primarily interested in having their problems solved and operations and competitive posture improved. They are not interested in producing some general knowledge that relates to some abstract problem. This means you get better access if you approach them with problem solvers (consultants) but you get better results if you include problem identifiers in the team (researchers). This is the way it works in medicine, and a similar logic can be applied in the social sciences as well. The reliance on actor-network theory, in turn, was necessary for rooting the solution deeply with the specific context and understanding the more tacit interdependencies between different factors affecting the logic of a business ecosystem. Figure 4 illustrates the iterative steps of the process. The dotted lines indicate situations where you revert back because the solution does not hold.

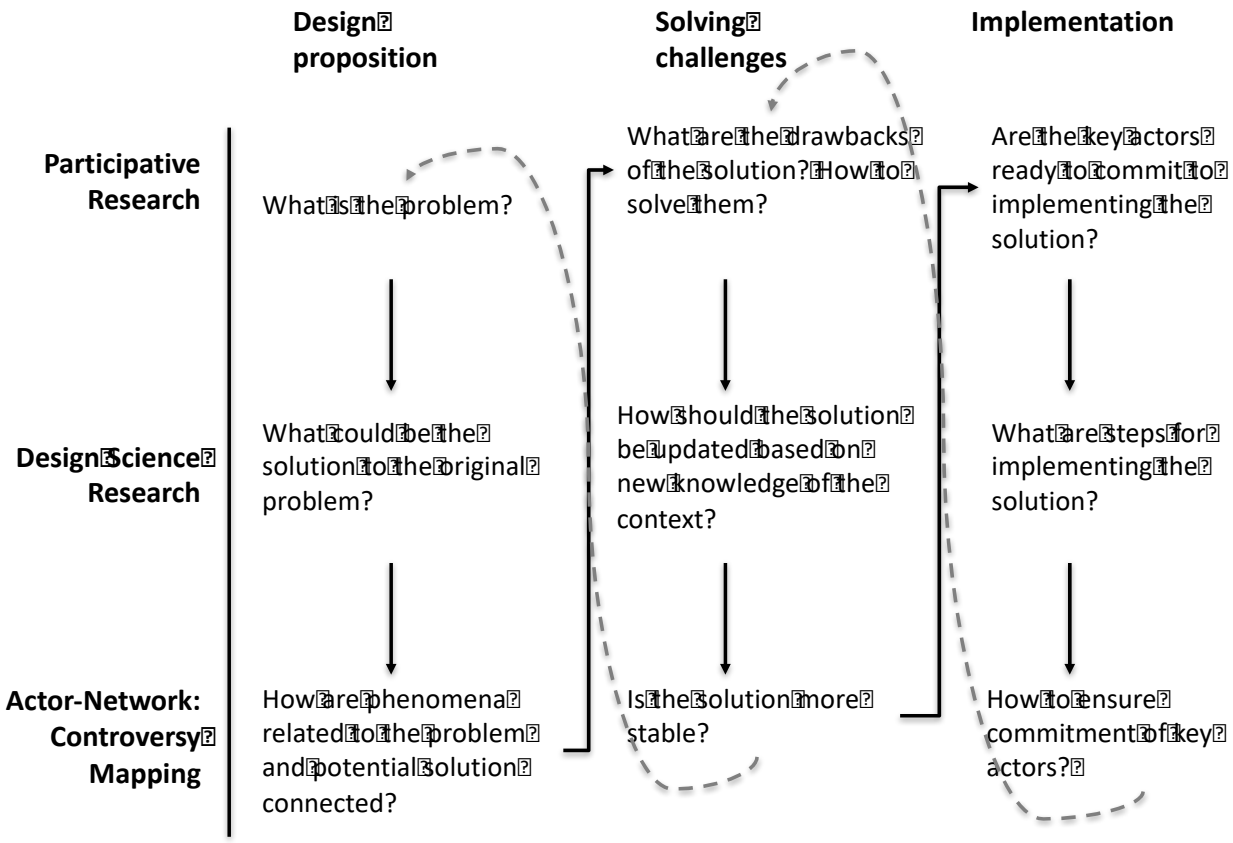


Figure 4 The epistemological steps of Transformative Business Studies

It is indeed more challenging to attempt to explore such wide research topics as design of business ecosystems, especially sustainable ecosystems. Choosing such a research goal, in turn, leads to research results that are difficult to formulate, present and transfer for practical use. However, the social significance of such results tends to be greater, which makes the journey worth the effort. Close collaboration between research and practice, conceptualization and application means we simultaneously transform the idea of how things work and what to do about it. By employing the transformative business research methodology presented in this paper, we believe, it is possible to conduct complex social and organizational research that can be systematically transferred to real-life applications and society's benefit. As stated in the beginning, transformative business studies focus on the real life challenges in society and finding out how they could be solved. Not all problems are related to business practices but business studies should try to identify the ones that can be solved through new business practices. This can be done by striving to solve the problems and identify what solves them and what does not and, based on this, contribute to the greater good of society by distilling that knowledge and making the knowledge transferable.

We recommend this methodology for researchers, practitioners and financiers. For researchers, basing the research on an existing practical problem can be rewarding both from a substance and a financial point of view. It also enhances theory development as theory this way is defined by the problem it solves or contributes to solving – the solution is in the problem. For a practitioner, transformative business studies bring a fresh perspective to the daily problems and allow to interact

with researchers committed to solving the problem by means that can be found in extant theoretical knowledge or that are yet to be discovered. For research financiers, it offers a better return on the money invested.

Since the method focuses on actual problems, it does not exclude any school of thought beforehand. We believe firmly that just as there are problems that are best solved through theories from medicine, technology or business studies, there are also those that can be solved with the help of theories found in the humanities. We propose that further research needs to focus on applying the methodology for transformative business studies in other contexts and further elaborating on methods related to it.

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